

10 Lessons On Platform Business Models With Platform Revolution Author - Sangeet Paul Choudary

Sangeet Paul Choudary is the best selling co-author of Platform Revolution and author of Platform Scale. Sangeet is also the young global leader at the World Economic Forum and founder at Platform Mission Labs and keynote speaker.

What actually drove you to this research into the platform business models?

Sangeet Paul Choudary: I got into platform business models quite some time back. I first started thinking about platforms back in 2006, 2007, when I was working at Yahoo and looking at new business models that we were investing in.

And one of the things that I was looking at was Alibaba. I realized when I looked at Alibaba back then that the way Alibaba thought of its business model was fundamentally different from the way Yahoo thought of its business model, which was a very linear way of an aggregate audience and serve them the ads.

And that's kind of got me thinking about what exactly was different about how Alibaba thought about this. I then moved on to a company called Intuit, and they started looking at how platform business models were fundamentally different from traditional software companies.

This was back in 2008 when data was still something that we had not figured out how to monetize. So a lot of these ideas were still pretty nascent at that point.

But sometime around 2011 or so, I condensed a lot of this thinking into this shift from **pipelines to platforms that essentially businesses were moving from the traditional linear business models, which I called pipelines to these new business models, which are platforms.**

Back in 2010, 2011, we had Uber, Airbnb, Facebook, Google, but we did not really see this as something much bigger than a Silicon Valley thing.

Today, we see this as applying to everything. When I came up with this formulation back around that time, I instantly saw that this could be applied across industries.

This was a fundamentally new way of thinking about how economic interactions can be structured, and that's what really got me excited.

I then quit my job, started doing this full time, collaborated with a group at MIT. We brought out a book, did a range of research and ever since then, it's spread across industries, across governments.

My work spans both helping businesses move in this direction, but also helping countries regulate platforms and come up with strategies to compete in the platform economy. So that's kind of a brief overview of my journey.

How do you think platform business models have impacted business strategy?

Sangeet Paul Choudary: The nature of competition has changed partly because the resources based on which we compete have changed.

The big shift that has really happened was that the resources on which we used to compete in the traditional industrial economy were supply-side resources that the company controlled either through ownership or through some form of contractual ownership, even if it was not within the boundaries of the company, of the firm.

What's changed is that as the world becomes more connected, whether you own the resource or not is secondary. **Whether you can understand the source and allocate the source is becoming the primary advantage.**

And so **data is becoming a very important asset because data helps you to identify, understand and allocate resources and then, in effect, control market interactions around that resource.**

In the traditional world, you had to control the resource, you control those market interactions, but now you can **just control those market interactions by managing data.**

And that is fundamentally what has changed because what then changes is that your competitive advantage is no longer constrained by what you have in-house. It can be any resource around which you can gather data and where you can aggregate and learn from that data and manage interactions around that resource.

And so because of this, fundamentally how companies compete have changed. Companies still compete because of access to resources and managing interactions around those resources.

But now, that can be done without actually owning those resources, that can be done through data. And that is fundamentally the biggest shift that has happened.

And this leads to a second factor that **when you don't own the resources, you can benefit from network effects because those resources lie outside your firm.**

And as you encourage more interactions around those resources, as third parties bring in their resources, you can then create a network of demand.

And as more demand comes in, even more, third parties come in with their resources. And that is what a network effect is.

So if you think of network effects, fundamentally hotels used to control the rooms that they would build, the rooms that they would serve, but Airbnb does not really control those resources in the same way.

And this is not to say that asset ownership is going away. Platforms will also continue to own the most critical assets that help them to compete.

So if look at Amazon, Amazon has to own the warehousing system because that is the critical asset that its ecosystem of third buddy merchants need from it.

And so I'm not trying to say that resources are going to go away, **but what's changing is that you can benefit from the sources that you do not control internally because you can now manage them to data.**

So it's a combination of the **digitization of resources coupled with** network effects that are fundamentally changing how competition works.

What strategic framework can we use to understand the competitive landscape today?

Sangeet Paul Choudary: So that's an interesting question because one thing that does change is that to assess the nature of competition, we also need to understand the nature of participation and collaboration.

Because when you do not compete based on your internal resources and your internal workforce only, you compete based on external resources and labor, then you need to think about what are the right incentives for these external parties to participate.

So the way you think about competition now, you need to start **first of all with identifying what the incentives for external producers and consumers to participate in my business model are.**

That is the first step. Now if you think of that question, that question is not a question about competition, but it's not too different from a question about how do we procure the resources on the basis of which we compete, which was what a traditional company would ask.

So today when you look at the competition:

- **The first thing you need to think about is the economic incentives for ecosystem participants.**
- **The second thing you need to think about is control points.** Once ecosystem participants participate, *what are the few factors that enable you to establish control over this ecosystem?*

This is where, again, if I take the Amazon warehousing example, Amazon creates incentives for users to come in and provides them Prime, it provides tools to the merchants.

But what helps it to hold both these sites together is the warehousing network. The warehousing network enables two-day delivery through Amazon Prime, and it enables Fulfillment by Amazon for third-party merchants who now don't need to do the fulfillment themselves.

So the control point is the warehousing network. So identifying the control point is the next important thing that you need to do once you created the incentives.

- **And the third thing is to think about scaling factors. What are the mechanisms by which your platform will scale?**

One scaling factor is what I talked about in the form of network effects. A second scaling factor is a learning effect, which today we think of as machine learning based on the more data you capture, the more you learn about interactions.

So **learning effects help you learn what is happening in the market.** So these are two important scaling factors that we need to think about when we think of competitive advantage.

To me, it's these three things that you need to think about.

1. Start with incentives,
2. create a control point
3. and then figure out how you scale both of these things through scaling factors.

What are the key differences from pipeline to platform business models?

Sangeet Paul Choudary: Well, there are three key differences between the pipeline and the platform business model.

- The first is that a **pipeline business model is just focused on serving its customers, its end consumers. A platform business model has to serve both sides, both contributors and producers in the business and end consumers.**

So one test that you need to look at is *do you actually have external parties contributing to your business model. If you don't, then there is no reason to call it a platform because a platform has to have external participation, external contribution.*

- The second key difference is whether you need to control the assets as I mentioned internally or whether you can manage interactions around assets and around resources, around users who are external to your system.

So that is where the key difference comes in. So if you were a staffing company or a recruitment company, you would control the temp staffing people you would provide.

But if you look at freelancing platforms like Upwork, they don't control those resources internally; they connect them to the client over the platform.

And similarly, you can apply that to the hotel example I gave into a whole range of other examples. **So that is the key difference, where do the assets sit.**

You could yourself control assets.

You don't have to be asset-light, but do you benefit from assets outside your system? That's the key part.

- And the third element is that a **pipeline business is very focused on product or service delivery.**

It's very focused on ensuring that the product is great, it's constantly shipping. A platform business, **the metrics it tracks are not internal. The metrics it tracks are about how third-parties are delivering products and services to the platform.**

Are they increasing their participation on the platform? So there's a big difference in the kind of metrics you track as well. So those are the key differences between a platform and the pipeline.

What are some of the metrics that platform business models need to track?

Sangeet Paul Choudary: The metrics that you should not be tracking just the size of your user base or just the number of users who are coming on board. Because very often, we think that platforms are about millions and billions of users and so we think that the proof of the platform is in the number of users coming on board.

But actually, that's not true!

What's important when you're building a platform is to **track what is the lead at which you are creating successful interactions – so** identifying that as the critical metric. So let's take an example.

If you launch a ride-hailing platform like Uber in a new city, and if somebody picks up their phone, a user picks up their phone and they open the app and they see no taxi available. That is a failure of a market interaction that you should be supporting.

So tracking what the success rate is, what is the failure rate of market interactions is one critical part, and you need to minimize that failure in market interactions.

And that is the reason why Uber then translates it into metrics for the drivers because drivers then are required to have a minimum acceptance rate.

If they get a certain number of requests for new rides, they cannot reject more than 10% of those requests. And that is to serve the overall metric of successful interactions eventually.

- **Metrics which focus on interactions are very critical to building platform businesses.**
- **A second metric that's very important as metrics that focus on scaling. How is the ecosystem scaling around you? Is the participation of a user increasing over time?** So think of Instagram. If a new user comes on board and they set up 20 photos in the first month, but they then move on to set up around 100 photos in the fifth month, that means that over time their participation on the platform is increased.

So look for indicators that show that the producers on your platform are increasing participation and so are the consumers. So those are the key metrics that are important when you're running a platform business.

Is there a difference between growth tools and network effects?

Sangeet Paul Choudary: Yeah. So that's a good question because there's a common misconception that people have between two things that sound very similar.

One is called network effects, and the other is called virality or viral effects. **So when you think about something going viral, people think of that also as a network effect.**

Now, the fundamental difference. **Virality is a phenomenon where the more people use your product or service, the more they spread the message about it.**

So if I take a few examples, the more I use Gmail, the more other people see that I'm getting they're receiving emails from Gmail. **The more they may want to switch towards Gmail over time.**

Initially, the more you would see photos from Instagram on Facebook, the more you would be exposed to Instagram.

So this is an example of virality where when you use the product, you spread the product externally to other parties.

Network effects are that more people using the platform create more value for others using the platform.

And so there's a difference.

Network effects create value on the platform. Viral effects spread the word about the platform or the product externally.

So network effects, an example is the more users who are on Airbnb. The more hosts are setting up listings on Airbnb, the more choice there is for travelers. Now that's a network effect.

Or take the example of YouTube, the more videos that are being set up on YouTube, the more choice I have as a viewer to view things on YouTube.

Now, if I take a video from YouTube and embed it on Facebook, that's not a network effect, that's a viral effect.

That is a good tool.

It's a mechanism to grow YouTube because people from Facebook then come back to YouTube. **So a viral effect is a growth tool that brings external users back to the platform. Whereas a network effect increases the value on the platform,** just like adding more than more and more videos onto YouTube.

What are network effects?

Sangeet Paul Choudary: This does get a little theoretical, but there are many different ways in which you should think about network effects.

- **The first is what is called a direct network effect,** which is users like you, the more they use a particular tool or service or platform, the more value you get out of it.

So if you think of the telephone that has a direct network effect, everybody else uses a telephone, the more people using the telephone, the more valuable it becomes.

- **There's a second thing which is a cross-sided network effect or two-sided network effect** where the more producers you have on a platform, the more consumers I attracted, and the more consumers you have, the more producers I attracted.

So the more hosts on Airbnb, the more value for travelers.

It's interesting to note that sometimes platforms which have this two-sided network effect actually have a negative direct network effect. So if you look at hosts, the more hosts that are in a particular city, the more competition there is for users among those hosts.

And so the direct network effect of having other hosts is negative, but the cross-sided network effect or the two-sided network effect of having other travelers is positive. So it's important to kind of think through these factors.

There are many other nuances around this. There's an idea of a local network effect where more users using a platform do not add value unless they're directly connected to you.

So think of Facebook, you really get value out of the users who you are directly connected to and the users, they are directly connected to.

The same thing applies to WhatsApp. So those are direct network effects, and those are local network effects.

And then there are different variations of utilities that have network effects to different degrees. So if you look at technical standards, they also have a network effect.

So if you look at a standard like the DVD standard or the CD standard, the more manufacturers who applied that standard, the more movie houses who then adopted that standard, the more users would then benefit from that standard.

So standards typically manifest network effects in much more complex B2B industry-level ecosystems.

So these are some of the ways in which you think about network effects. There's a lot that's been written about 10 or 15 or 13 different types of network effects.

But fundamentally, the most important thing to understand is that there's a difference between who is creating value with its users who are just like you or users who are consuming from you or users who you are consuming from. The one-sided versus two-sided network effect is the single biggest difference to look at.

Apart from that, a couple of other things to look at are the strength of the network effect.

What exactly leads to a stronger network effect and does the network effect stop increasing after a certain point?

So if you look at Uber, for example, there is potentially some level of a network effect initially where the more drivers come in, the more likely it is that you find the ride.

But beyond the point, including the number of drivers does not improve your experience. Once you reach a certain level of waiting time beyond that, more drivers coming in, in a particular city does not improve your waiting time any further.

And so the network effect, the value of it kind of does not increase beyond a certain level. *But there are others, platforms which enable the long tail like eBay whereas you keep on adding more than more variety of sellers, you keep creating more and more choice.*

And so you need to think about at what point the value of the network effect is leveling out or whether it's still doing beyond that point.

That's a critical thing to think about.

What are negative network effects? And why it is important to understand them?

Sangeet Paul Choudary:

So a negative network effect essentially means that the more the network grows, the value for the users goes down.

And in the traditional world, this used to be called congestion.

So the more people using the highway system, the more traffic jams you end up in. Or the more people in a room, the less likely it is to have good decent conversation just because it gets crowded, but also because everybody is talking too loudly and so you can't hear and you can't meet the right person within that room.

So we understand congestion in traditional terms because in the traditional world, we have networks that were limited by scale.

The scale of a highway network or the total scale of loads in a certain city or the total size of the room or of a conference hall, these are all limited.

But in the online world, in the digital world, the scale limitations of the underlying infrastructure go away. And that is why we start talking more about positive network effects.

But negative network effects come up even in the online world. Not so much because of scale limitations, but because of limitations in quality management or curation.

Because what happens is the more users come on board, the more difficult it becomes to manage quality of the interactions.

So unless you set up curation mechanisms upfront, it becomes difficult to ensure that the quality is managed on a consistent basis as you move forward, and as you scale even further.

This is where negative network effects become important because the more users that keep coming on board, if your quality control mechanisms are not in place and more importantly are not scaling at the rate of the network, then the quality dips and then people start seeing less value and start leaving the network.

And this happens from time to time on various kinds of networks and that's the negative network effect that we see.

So it's important to make the distinction between congestion that happens because of scale limitations and noise that happens because of curation limitations.

So those are some ways in which negative network effects manifest.

What is the chicken and egg dilemma? Why is that important?

Sangeet Paul Choudary: The chicken and egg dilemma happens because network effects essentially mean that the value is being created by our users. So when there are no users, there is no value.

So then you have this chicken and egg. If there are no producers of value on your platform, consumers won't come. And if consumers don't come, producers wouldn't want to come.

So who do you get first? The chicken or the egg? The producer or the consumer?

That is the key dilemma that comes up.

This is a classic dilemma that is faced by all platforms which rely on the ecosystem to create value. There are many different ways of solving the chicken and egg problem.

I actually go into this in a lot of detail in my books as well as in my online course around platforms. And there are many different ways in which this dilemma manifests itself.

But fundamentally, there are two ways in which you solve this:

- The first way to solve it is that **you create some value in your platform even without the networks.** So you create some value within the tools and the software itself. An example of this is Open Table. *So OpenTable today, the network that connects restaurants with consumers, but initially it launched as a restaurant management software, which could work even if consumers were not onboard. So the restaurants would still find value in using it.*
- The second is to **ensure that you launch the platform in a high activity market.** So think of Facebook launching as a closed system inside Harvard University. *High activity, everybody knew each other, everybody came on Facebook and started interacting.*

Think of Twitter launching at South by Southwest and people started tweeting about the conference and everybody else saw the tweet. So there was a lot of interaction. Think of PayPal launching as a payment mechanism on eBay where transactions are already happening; they just needed a payment mechanism.

So **successful platforms always find these high hubs of activity,** whether they are other platforms as PayPal with eBay or whether they are physical locations like Facebook or whether they are events like Twitter.

All of them always look to find these high activities, high concentration of activity as an initial way to kickstart the network effect.

Where should you build a platform business model today?

Sangeet Paul Choudary: The basics of business do not change. You need to start with a clear user pain point. You don't build a platform because it's a good thing to build or because that's the way business is headed, or because they are attracting high valuations.

You start with a user pain point, that is the starting point and you try to see what is the best way of solving that user pain point. And in doing that, you might realize that you need a platform model to solve the user pain point.

So the way to build a platform, again, starts with the user pain point itself.

And then there might be many ways of solving it and the platform might be one way of doing it.

Now, once you got into a point where you realize that a platform is a right way to do it, then you need to figure out who are you primarily solving this problem for.

Is it the consumer side of the producer side?

So if you ask me if Facebook is primarily solving a problem for the users, not for the brands. The brands come on board later. The media companies come on board later. The advertisers come on board later.

So the primary user, in that case, is the consumer, but if you look at Open Table, I would argue that the primary problem is being solved for the restaurant because it is the restaurant management, seating management. And then as an additional kicker, the restaurants get connected to the user as well.

So always look for who's the primary party you're solving the problem for, and then try to focus on uniquely solving that problem especially well for that primary body first before you start optimizing the experience of the secondary party.

That is broadly how I would think about solving this or getting started because you need to balance incentives on an ongoing basis. But initially, you need to know for sure which side producer or consumer is driving value creation.

In the case of Facebook, consumers drive engagement and data creation. In the case of Open Table, the restaurants create data about which seats are available.

So who is driving that value creation on your platform? You need to identify that before you set about solving this problem uniquely.

What are you working next?

Sangeet Paul Choudary: I look at my work in terms of two models in terms of what's next to be researched, but also how can I have an impact with the work that I'm doing.

In terms of research, I'm looking at a few specific elements. I'm looking more at antitrust, but just because I also get asked a lot more of those questions around how to think about monopoly and antitrust in the case of platforms.

The second thing I'm looking at is global trade, how does trade change and how do countries compete in a platform economy, what will global trade look like when the world moves towards platforms and ecosystems.

And I've worked with quite a few countries helping them with their trade strategy.

The third thing I'm looking at is B2B ecosystems. What does it look like? How is that different from the B2C examples we talked about over here? What does it mean to create a logistics ecosystem or a financial services ecosystem to serve the unbanned? Or how do you change the surge in clinical trials are conducted by pharmaceutical companies today? How do you think of that in a platform economy?

So I'm looking a lot at understanding competition, commoditization, new business models, further up the supply chain. In terms of the impact of all of this, there are different ways in which I've been scaling my work. I've done a lot of advisory work, high-level advisory work with companies and governments.

I try to scale it further through my writing but also through creating educational products, which I licensed to firms.

And then I also have been investing and creating ventures for corporates who are looking to build and compete in this platform economy.

So it's a combination of different models through which I try to apply and scale the impact of what I work on.

Where can people find valuable resources to understand platform business models?

Sangeet Paul Choudary: I've created quite a lot of resources on this topic. I've written two books, Platform Revolution and Platform Scale, which have both been bestsellers. So they are both good starting points to understand platform business models.

You also have my website, platformthinkinglabs.com, where there are a lot of resources, including different articles. There are videos on YouTube as well.

And I've launched an enterprise course which essentially enterprises licensed to have their entire teams learn about platform business models.

So if you're working at a company and if you need to understand or have your whole team understand different aspects of platform business models, that's of course that you could consider looking at as well.